

Auteur: Rob Willemsen
Studierichting: Informatica
Oefeningenreeks 4A nr. 8.11

$$\int \frac{dx}{\tan(3x+5)}$$
$$= \int \frac{\cos(3x+5)}{\sin(3x+5)} dx$$

stel $t = \sin(3x+5)$, dan is $dt = 3 \cos(3x+5) dx \Rightarrow \frac{1}{3} dt = \cos(3x+5) dx$

$$= \int \frac{\frac{1}{3}}{t} dt = \int \frac{1}{t} \cdot \frac{1}{3} dt = \frac{1}{3} \ln |t| + C$$
$$= \frac{1}{3} \ln |\sin(3x+5)| + C$$

References